

**Final Project Report**

**ME4311**

**Micro/Nano Electromechanical Systems and Nanotechnology**

# **Design of a MEMS Pressure Sensor Array for Real-Time Gait and Posture Monitoring in Wearable Footwear**

**By**

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# 1 Introduction

There are different human movements that generate complex patterns of plantar pressure across the foot sole. Therefore, measuring these pressure values accurately is important in gait analysis, posture assessment, diabetic foot monitoring, footwear design, rehabilitation, and sports biomechanics.[1] Abnormal pressure distribution is obtained with gait instability, foot deformities, and a high risk of diabetic ulceration conditions that represent significant medical and economic burdens worldwide. Studies report that diabetic foot ulceration alone costs over \$1 billion annually in the United States.[2] Therefore, reliable, high-resolution, real-time plantar pressure sensing system is a critical technological requirement in modern biomedical and wearable systems.

Conventional plantar pressure measurement systems are resistive, capacitive, or conductive polymer-based sensors. While these systems are widely used in platforms and in-shoe configurations, they suffer from several limitations, including hysteresis, nonlinearity, creep, thickness constraints, and limited pressure ranges.[3] Previous research shows commonly used resistive sensors consist significant hysteresis and inaccurate repeatability under dynamic loading conditions, capacitive sensors often require thicker dielectric layers, reducing flexibility and wearability. These problems limit the suitability for continuous gait monitoring, during high-pressure loading conditions observed during walking, running, or jumping.[3]

Recent studies show that MEMS pressure sensors offer significant improvements over traditional sensors because of their miniaturization, superior mechanical properties, low power consumption, and high linearity under large pressure ranges.[3] MEMS piezoresistive pressure sensors have emerged as a leading candidate due to their high sensitivity, low hysteresis, excellent temperature stability, and capability to withstand pressures exceeding 1–2 MPa[2], making them ideal for foot plantar applications where peak pressures can reach Existing Foot Pressure Measurement Technologies 1900 kPa.[2] The diaphragm-based structure on these enables precise stress concentration, which can be controlled through careful placement of piezoresistors on high-stress regions of the membrane, as demonstrated in multiple optimization studies.[2]

Furthermore, MEMS technology enables highly scalable sensor arrays, essential for accurately mapping spatial pressure distribution across the foot.[4] Prior works have explored customizable or trim-to-fit sensor arrays for gait analysis however, most commercially available systems rely on polymer-based pressure fabrics with limitations in durability and

accuracy under high load conditions. By integrating silicon-based MEMS piezoresistive sensors into a wearable footwear platform, it becomes possible to obtain high-resolution, real-time pressure data with enhanced reliability, robustness, and long-term stability.[4]

Given these motivations, this project focuses on the design, simulation, and optimization of a MEMS piezoresistive pressure sensor array tailored for real-time gait and posture monitoring in wearable footwear. Leveraging silicon diaphragm sensor theory, Wheatstone bridge configurations, piezoresistor placement optimization, and finite element analysis, the project aims to create a novel array architecture that addresses key challenges in existing systems, namely high-pressure tolerance, accurate load transfer through footwear materials, and minimization of mechanical crosstalk between sensing cells.

## 2 Literature Review

### 2.1 Plantar Pressure Measurement and Gait Biomechanics

#### 2.1.1 Gait cycle and plantar pressure distribution

Plantar pressure describes the distribution of force applied by the foot to the ground while standing, walking, and running. It is a critical biomechanical metric used in clinical diagnostics, rehabilitation, orthotics design, sports science, and diabetic foot ulcer prevention. Understanding plantar pressure patterns enables early detection of abnormalities in gait and postural control.

The human gait cycle consists of stance and swing phases, with plantar pressure being primarily captured during the stance phase. Studies consistently show that pressure distribution is highly nonuniform, with peak forces concentrated at the heel, metatarsal heads, and hallux (big toe).[5]

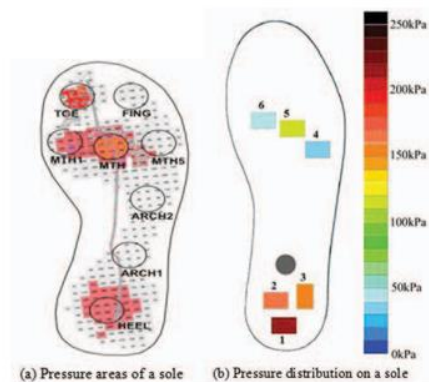


Figure 1: Pressure area of a sole[5]

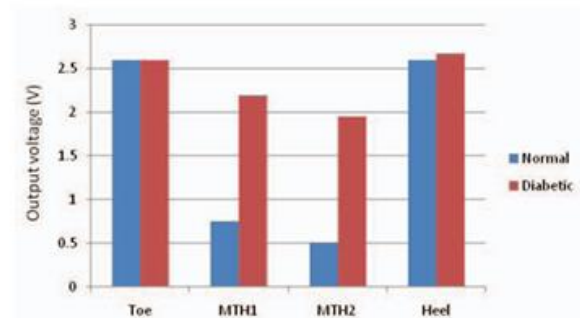


Figure 2: Difference in foot pressure of normal and diabetic patient[5]

In Figure 1 shows major pressure zones of the foot during typical stance and a heat-map-style pressure plot. This outlines,

- Heel region exhibits the highest peak pressure[5]
- Metatarsal heads (MTH1–MTH5) show moderate pressures[5]
- Toe regions exhibit variable pressures depending on foot type[5]
- Arch area typically exhibits minimal pressure[5]

This aligns with clinical studies showing that maximum pressures typically occur at heel strike and push-off phases.[5]

#### Pressure Values

In foot pressure studies, pressures commonly range between,

- **Walking** - 1000 kPa[2]

- **Running** - can exceed 1 MPa[2]
- **Diabetic patients** - abnormally high pressures at MTH regions[5]

The importance of accurate high-range measurement is strengthened in sensors who note sensors should withstand pressures up to 2 MPa[2] for biomechanical applications.

### 2.1.2 Applications of plantar pressure analysis

#### Clinical Diagnosis & Diabetic Foot Assessment

Abnormally high peak pressures under metatarsal heads are strongly associated with ulcer formation.

Figure 2 shows bar-chart comparisons of normal vs diabetic plantar pressure, clearly demonstrating increased pressure at MTH1 and MTH2 in diabetic patients.

This shows that diabetic feet consist of a higher forefoot loading, confirming plantar pressure as a diagnostic biomarker.

#### Balance and Fall-Risk Assessment

Figure 3 shows plantar pressure arrays help evaluate human balance stability by monitoring pressure shifts across the foot.[6]

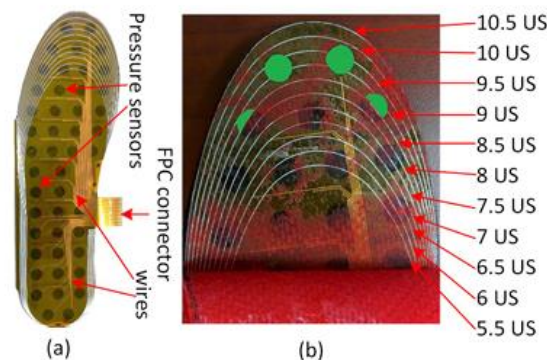


Figure 3: Insole shaped customizable pressure sensor array for different shoe size[6]

#### Sports Science[2]

Plantar pressure assists in analyzing running mechanics, impact force distribution, and overuse injury risk. In research it shows sports activities require sensors with higher tolerance due to peak pressures exceeding walking thresholds.[2]

#### Prosthetics and Orthotic Design[7]

Pressure mapping helps optimize load distribution inside prosthetic sockets and custom insoles.[7]



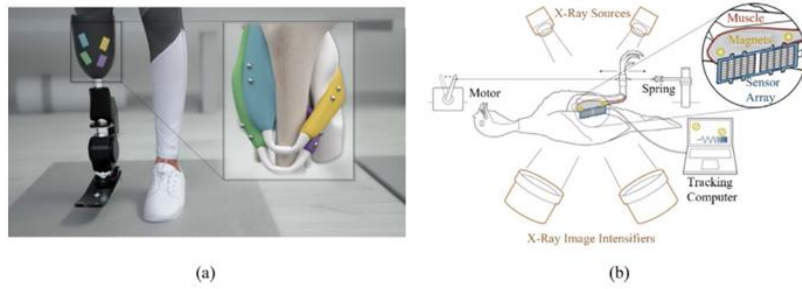


Figure 4: Wearable sensor application[7]

## 2.2 Existing Foot Pressure Measurement Technologies

Plantar pressure measurement systems have been widely adopted in clinical gait analysis, sports biomechanics, prosthetics, rehabilitation, and diabetic foot ulcer prevention. These systems aim to quantify the distribution of pressure between the foot and the ground or footwear during dynamic movement. Existing technologies can be broadly categorized into platform-based systems, in-shoe sensing systems, and emerging textile/printed sensor arrays. Although these methods provide valuable insight into gait mechanics, each comes with limitations in spatial resolution, durability, accuracy, linearity, and pressure range factors highly relevant to wearable continuous monitoring.

### 2.2.1 Platform-Based Pressure Measurement Systems

Early plantar pressure systems used pressure platforms such as Novel and Tekscan.[3] These sensors include hundreds to thousands of pressure-sensing elements embedded in a rigid plate. It allows high-resolution static and dynamic pressure mapping. In Table 1 it shows that these platform devices are still widely used because they offer,



Figure 5: Platform-based foot plantar pressure sensor[3]

- High spatial resolution
- High sampling frequencies
- Reliable calibration stability

However, they also require subjects to step accurately in the sensing region, causing gait adjustments and limiting ecological validity. They cannot be used inside footwear and are therefore unsuitable for long-term monitoring or outdoor activity.[3]

Table 1: Technical specifications of some existing sensors and 2 different MEMS proposals[3]

|                         | Vista Medical<br>[44] | Tekscan<br>[26] | Novel<br>[24] | Parotec<br>[45]                   | Textile Sensor<br>[30] | Wahab <i>et al.</i><br>[43] |
|-------------------------|-----------------------|-----------------|---------------|-----------------------------------|------------------------|-----------------------------|
| <b>Sensor Size</b>      | 2 mm thick            | 0.15 mm thick   | 1.9 mm thick  | ~4 cm <sup>2</sup><br>(hydrocell) | Not Specified          | 2 mm thick                  |
| <b>Number of Sensor</b> | 128 (in shoe)         | 960 (insole)    | 99 (insole)   | 24 (insole)                       | 6 (insole)             | 15 (insole)                 |
| <b>Range (kPa)</b>      | 260                   | 1,034           | 1,200         | 625                               | 800                    | 3,000                       |
| <b>Frequency (Hz)</b>   | Not Specified         | 500             | Not Specified | 250                               | 100                    | 200                         |
| <b>Hysteresis</b>       | Not Specified         | 24%             | <7%           | 0.05% at<br>20 Ncm <sup>-2</sup>  | Not Specified          | Negligible                  |

### 2.2.2 In-Shoe Pressure Measurement Systems

To capture pressure during natural activity, in-shoe systems were developed. These embed flexible sensors inside insoles, making them suitable for sports applications, diabetic monitoring, and ambulatory gait studies.

#### (a) Resistive Force-Sensing Resistors

Resistive sensors, such as Tekscan's F-Scan system[3], operate using conductive polymer layers whose electrical resistance decreases under applied load. The sensing structure consists of two Mylar sheets coated with conductive ink, forming a variable-resistance interface that changes proportionally with plantar pressure. These sensors provide several advantages, including high flexibility, a low-profile form factor, and the ability to capture temporal gait parameters such as center of pressure, step time, and loading rate.[3]



Figure 6: Foot plantar pressure sensor F-Scan[3]

#### (b) Capacitive Sensors

Capacitive sensors, which are pressure sensors, are based on the principle that capacitance changes between two metal electrodes when separated by a dielectric layer elastic layer, where pressure compresses the dielectric and changes the distance between the electrodes. An example of such technology is the Novel Pedar and Novel EMED pressure platform, which are commercial applications of this technology. Pedar sensors are known to possess very good linearity and low hysterical values, unfortunately, the thickness limitation can be posed by the

multilayer dielectric structure, and thus it requires proper calibration to stay on track with the accuracy. Even though capacitive sensors outperform resistive ones, they also have a few shortcomings, such as a limited pressure range, a decreased response to repeated flexion during gait, and increased costs to manufacture because of complex construction.

### 2.2.3 Textile- and Fabric-Based Sensor Arrays

More recent developments have presented piezoresistive fabrics as cheap, extremely stretchable substitutes of wearable measurement systems of plantar pressure. A flexible, customizable in-shoe array of pressure sensors made of EeonTex prescribed piezoresistive material between patterned copper pads on flexible PCBs.[6] The piezoresistive cloth has pressure-sensitive resistance, where it is possible to create a number of discrete sensing cells. Their system allows trimming into different shoe sizes, and the signal scanning is done with sequential multiplexers using voltage-divider circuits and 48 equally spaced sensing points in a shape that fits a complete insole. It also showed that insole trimming did not have any influence on gait-cycle or cadence detection, showing the suitability of the system to detect temporal gait. They however also measured variability as high as 8% in the uniformity of materials in the fabric and resistive textile sensors are typically characterized by low accuracy, nonlinearity, row-column addressing crosstalk, and drift. These limitations go a long way to reduce their usability in clinical-grade plantar pressure measurement.[6]



Figure 7: The consistency test of the piezo-resistive material. The numbers from 1 to 9 indicate the locations of the testing points on the material.

Table 2: Results of Consistency Testing

| Position number        | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    |
|------------------------|------|------|------|------|------|------|------|------|------|
| Resistance( $\Omega$ ) | 50.3 | 52.0 | 46.8 | 50.3 | 45.9 | 45.1 | 48.0 | 51.8 | 50.7 |
| Variation(%)           | 2.7  | 6.2  | 4.4  | 2.7  | 6.2  | 8.0  | 2.0  | 5.6  | 3.4  |

## 2.3 Limitations of Current Foot Pressure Sensing Systems

Despite the existing diversity of technologies to measure the plantar pressure in the categories of resistive, capacitive, piezoelectric, textile-based, and platform-mounted sensors, none of them can be utilized in wearable and real-world environments due to inherent limitations that diminish the accuracy of measurements, its reliability and durability. These are serious

limitations of their application in high precision gait and posture observance and explain why more robust and high-resolution devices like MEMS-based sensor arrays are necessary.

### **2.3.1 Hysteresis and Signal Drift**

One of the most nagging issues in the resistive and textile pressure sensors is hysteresis. The F-Scan which is an in-shoe resistive platform used by the company Tekscan has a system with as high as 24% levels of hysteresis that is, the unloading curve does not follow the loading curve even at the same level of pressure.[1] This hysteresis can also result in challenges of getting correct force values when transitioning between fast gaits. Drift and creep are also issues of concern. The gradual variations in output of resistive and textile sensors under constant loading result in the inaccurate long-term output of such sensors, which are especially problematic in continuous gait measurements used in footwear. Also, the consistency of gait-cycle detection, absolute pressure values changed over time because of the instability of the materials in piezoresistive fabrics.[6]

### **2.3.2 Limited Pressure Range and Sensor Saturation**

Most of the commercial in-shoe systems work within small limits of pressure. Yet, plantar pressure during the performance of such tasks as running, jumping, abnormal gait may exceed 1900 KPa.[2] The result of this difference is sensor saturation, lower measurement accuracy, and clipped measurements in high-force systems. Conversely, MEMS sensors can withstand up to 3000 kPa with a small hysteresis and this demonstrates the discrepancy between existing in-shoe systems and the demanded levels of biomechanical performance.

### **2.3.3 Spatial Resolution Limitations**

Systems of commercial platforms offer thousands of sensing elements, although such high resolution is at the price of poor practical usability, because the user must step on the platform exactly. In-shoe systems generally contain 99-256 sensors, much less than platforms, restricting its detection capability to local elevated areas of pressure, Development of diabetic ulcers at an early stage, Minor deviations of lateral gait, forefoot and toe abnormalities. It is observed that systems with only 47 discrete sensors are not detailed enough to map the whole plantar.

### **2.3.4 Material Non-Uniformity and Manufacturing Variability**

Although it is cheap and flexible, textile pressure sensors exhibit large variation in material characteristics. A study of piezoresistive fabric resistance under the same loads over nine test points showed up to 8% variation in piezoresistive fabric resistance.[6] This variation has a direct influence on the calibration accuracy. In addition, it is convenient but may also change the sensitivity of fabric sensors as the size of the sensors is trimmed to fit the size of the shoe, unless this is done carefully. Non-uniformity in manufacturing is also an issue that impacts performance in resistive sensors, where layers of conductive ink become damaged over time as the sensor gets bent, gets wet, and has sweat.

## 2.4 MEMS Pressure Sensor Types and Existing Designs

One of the most developed and thoroughly commercialized categories of MEMS devices are micro-Electro-Mechanical Systems pressure sensors which find wide and common application in automotive, biomedical, consumer electronics and industrial systems. The article reviewed gives a comprehensive categorization in MEMS pressure sensor designs, operating principles, structural designs, fabrication demands and performance trade-offs. These insights are synthesized in this literature review with regards to biomedical and wearable sensing applications including plantar pressure monitoring.

### 2.4.1 Classification of MEMS Pressure Sensor Designs

The MEMS pressure sensors can classify into four fundamental types based on their transduction mechanism,

- Capacitive MEMS pressure sensors
- Piezoresistive MEMS pressure sensors
- Piezoelectric MEMS pressure sensors
- Optical MEMS pressure sensors

Each category exhibits unique advantages and limitations determined by materials, structural configurations, and fabrication processes.

#### 2.4.1.1 Capacitive MEMS Pressure Sensor Designs

These sensors operate by measuring the change in capacitance between a deformable diaphragm and a fixed electrode. These sensors typically use,

- A silicon or polysilicon diaphragm
- A vacuum cavity created through bulk or surface micromachining
- Metal electrodes patterned on top and bottom surfaces

#### Working Principle

Capacitance is inversely proportional to the electrode gap

$$C = \epsilon \frac{A}{d}$$

When pressure deflects the diaphragm, the separation distance  $d$  decreases, increasing capacitance.

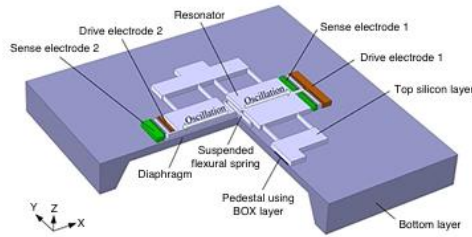


Figure 8: Capacitive pressure sensor design

### Advantages

- Excellent linearity at small deflections
- High sensitivity
- Low temperature dependence

### Limitations

- Sensitive to parasitic capacitances
- Require complex readout circuitry
- Limited compatibility with high-pressure environments

For wearable plantar pressure sensing demands a large pressure range. Therefore, capacitive sensors are less favorable.

#### 2.4.1.2 Piezoresistive MEMS Pressure Sensor Designs

The piezoresistive pressure sensor category is the most widely used and most relevant for high-pressure biomedical applications such as gait and posture monitoring.

### Working Principle

Piezoresistive sensors rely on changes in resistivity of doped silicon resistors placed on a flexible silicon diaphragm. When stress is induced,

$$\frac{\Delta R}{R} = \pi_l \sigma_l + \pi_t \sigma_t$$

The resistors are arranged in Wheatstone bridge, enabling differential voltage measurement.

### 3.2 Structural Designs

The article shows several common membrane configurations:

#### (a) Square Diaphragm[8]

- Most widely used
- Uniform stress distribution

- High sensitivity and simple fabrication

**(b) Circular Diaphragm[8]**

- More uniform radial stress
- Slightly reduced sensitivity compared to square membranes

**(c) Bossed Diaphragm**

- A thick central region improves stress concentration at the edges
- Enhanced sensitivity at higher pressures

**(d) Quad-Island or Multi-Island Designs[8]**

- Thin flexible islands improve membrane compliance
- Higher displacement at lower pressures
- Useful in low-pressure biomedical sensors

**Advantages**

- High sensitivity
- Simple and stable signal conditioning
- Excellent performance under high pressure
- Mature fabrication processes

**Limitations**

- Temperature dependence requires compensation
- Mechanical stress must be carefully engineered
- Diaphragm non-uniformity affects accuracy

These designs use a piezoresistive diaphragm for a MEMS pressure sensor array.

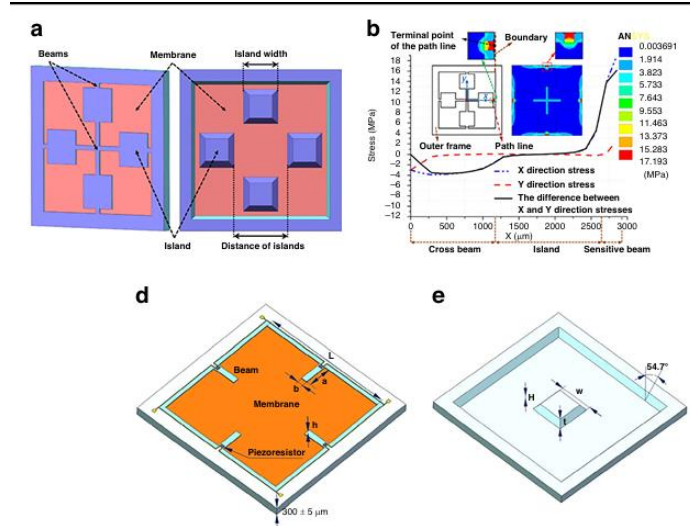


Figure 9: Diaphragm designs[8]

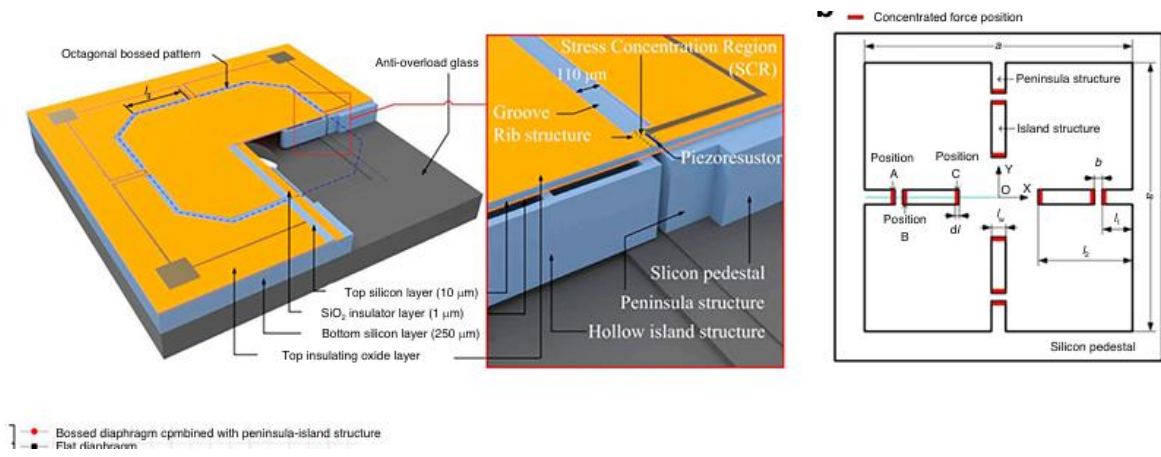


Figure 10: Piezoresistor position[8]

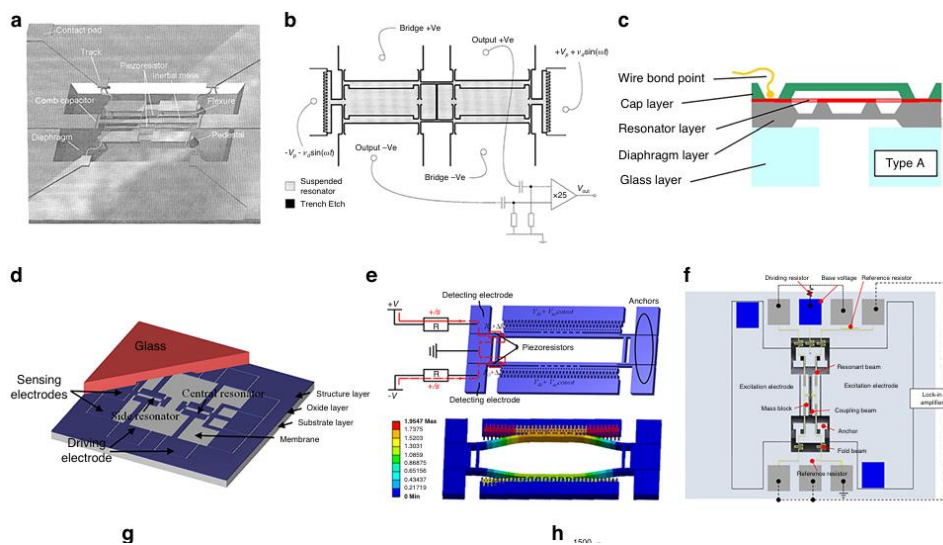


Figure 11: Exploded view of piezoresistor[8]



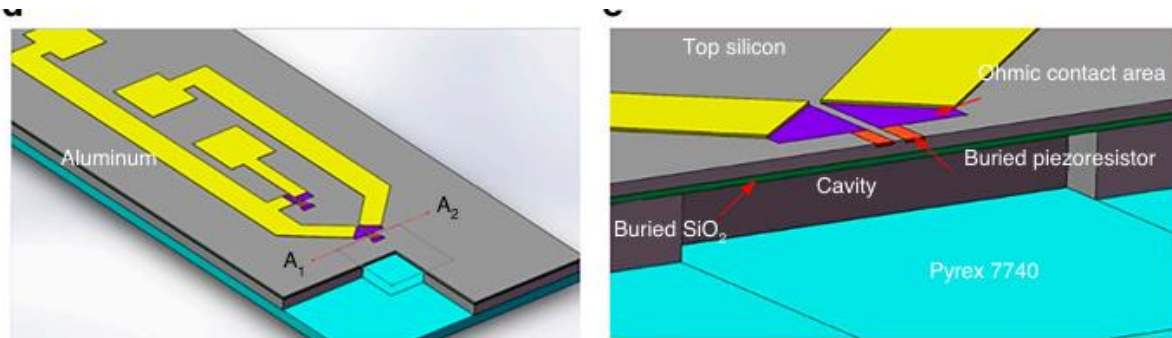


Figure 12: Type of piezoresistor design[8]

#### 2.4.1.3 Piezoelectric MEMS Pressure Sensor Designs

The article discusses piezoelectric sensors using PZT, AlN, or ZnO thin films deposited on silicon diaphragms. When pressure is applied, these films generate electrical charge.

##### Design Types

- Diaphragm-based piezoelectric structures
- Cantilever-based pressure sensors
- Bulk-mode resonant structures

##### Advantages

- No external bias voltage needed
- Very fast response
- Suitable for dynamic pressure measurement

##### Limitations

- Inability to measure static pressure accurately
- Charge leakage over time
- Poor suitability for gait/posture monitoring

Thus, while useful in microphones and impact sensors, piezoelectric MEMS sensors are not ideal for continuous plantar pressure measurement.

#### 2.4.1.4 Optical MEMS Pressure Sensor Designs

Optical MEMS pressure sensors rely on,

- Fabry-Perot cavities
- Fiber Bragg gratings
- Interferometric measurement

The article presents designs using silicon membranes with integrated optical cavities.

### Advantages

- Immune to electromagnetic interference
- High sensitivity and resolution

### Limitations

- Complex fabrication
- Bulky optical readout systems
- Difficult integration into wearable insoles

Thus, optical MEMS is unsuitable for your application.

*Table 3: Pressure sensors comparison*

| Sensor Type         | Sensitivity    | Static Pressure Capability | Power Requirement | Complexity     | Suitability for Plantar Pressure |
|---------------------|----------------|----------------------------|-------------------|----------------|----------------------------------|
| Capacitive MEMS     | High           | Medium                     | Low               | High (readout) | Moderate                         |
| Piezoresistive MEMS | Very High      | High (>1 MPa)              | Moderate          | Low            | Excellent                        |
| Piezoelectric MEMS  | High (dynamic) | Low                        | None              | Moderate       | Poor                             |
| Optical MEMS        | Very High      | Medium                     | High              | Very High      | Poor                             |

Piezoresistive MEMS sensors clearly provide the best balance of sensitivity, linearity, and high-pressure capability.

Table 4: Comparison of pressure sensor types

| Pressure Sensor Type                                 | Principle                                                                                                                                  | Advantages                                                                                                                 | Disadvantages                                                                     |
|------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Piezoelectric Pressure Sensor [30][34][50]           | When the asymmetric material is stimulated by an external force, internal polarization and surface charge occur.                           | High sensitivity; fast response;<br>Simple structure; high signal-to-noise ratio;<br>Independent of external power supply; | Limited static-force measurement;<br>presence of pressure drift;                  |
| Piezoresistive Pressure Sensor [19][50]              | When the monocrystalline silicon material is subjected to an external force, the resistance changes in response to mechanical deformation. | Simple operation; low cost;<br>high-frequency response;<br>suitable for dynamic and static measurement;                    | Low signal-to-noise ratio;<br>poor temperature performance;                       |
| Capacitive Pressure Sensor [19][38][50][51]          | When the distance between electrodes is changed by an external force, the capacitance changes.                                             | Good repeatability; simple structure;<br>Low power consumption;<br>good dynamic performance;                               | Low sensitivity; high cost;<br>complicated structure;<br>sensitivity to humidity; |
| Inductive Pressure Sensor [40][19]                   | When the sensor is subjected to an external force, the detection coil shifts, and the voltage between excitation coils changes.            | High resolution; high sensitivity;<br>low cost; low weight;<br>extension capability;                                       | Large size;<br>small measuring range;                                             |
| Crack-based Pressure Sensor [41][52]                 | When the crack-based sensor is bent under pressure, the nano-cracks disconnect, and the resistance increases.                              | High sensitivity;<br>Easy fabrication;<br>high repeatability; high flexibility;                                            | Hysteresis occurs at high strains;                                                |
| Triboelectric Nanogenerator (TENG)-based Sensor [42] | When a contact-separation event happens between two triboelectric layers, the mechanical triggering is converted into electrical output.   | Easy fabrication; fast response;<br>High durability; excellent robustness;<br>high sensitivity;                            | /                                                                                 |

### 3 Proposed Design

The objective of the proposed system is to come up with an MEMS pressure sensor array with a high-accuracy high-range suitable in real-time gait and posture sensors in wearable footwear. Current in-shoe technologies such as resistive FSRs and textile sensors have weaknesses in hysteresis, creep, drift, low accuracy and lack of pressure range under 150-600 kPa. In comparison, it might be that during pathological gait, walking or running, plantar pressure may be more than 1-2 MPa, which may need a sensing technology that can accommodate large loads with high repeatability.

To overcome these limitations, the proposed design uses,

- A piezoresistive MEMS diaphragm sensor cell optimized for MPa-range pressures
- A multi-cell MEMS sensor array placed according to plantar anatomical pressure hotspots
- Independent diaphragm islands separated by isolation trenches to prevent mechanical cross-talk
- A mechanical coupling layer that transfers foot pressure to each MEMS diaphragm
- On-chip Wheatstone bridge

#### 3.1 Single MEMS Pressure Sensor Cell Design

##### 3.1.1 Diaphragm geometry

The diaphragm is designed based on classical thin-plate bending theory, which describes the deflection behavior of a square membrane under uniform pressure loading. According to Timoshenko's small-deflection model[9], the maximum diaphragm deflection is given by,

$$w = \frac{0.0159 p(1 - \nu^2)A^4}{EH^4} [9]$$

where:

- $p$ = applied pressure,
- $\nu$ = Poisson's ratio,
- $E$ = Young's modulus of silicon,
- $A$ = side length of the diaphragm,
- $H$ = diaphragm thickness.

This relationship shows that diaphragm deflection is extremely sensitive to its thickness, varying with  $1/H^4$ . For linear and stable sensing performance, small-deflection theory requires the maximum displacement to remain below 3% of the diaphragm thickness.[9]



Figure 13: Side view of a thin diaphragm

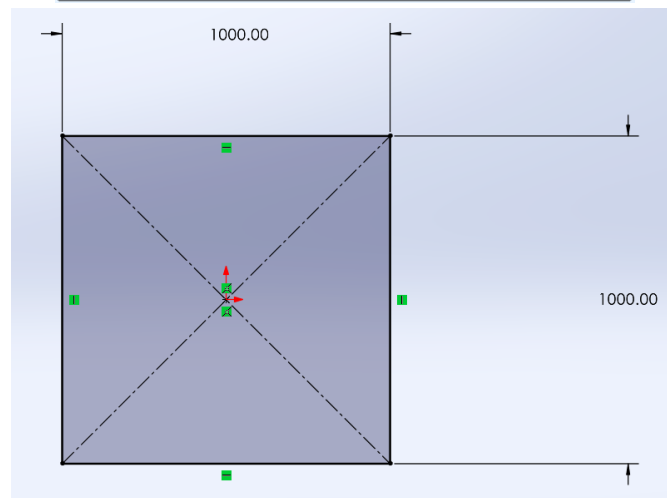
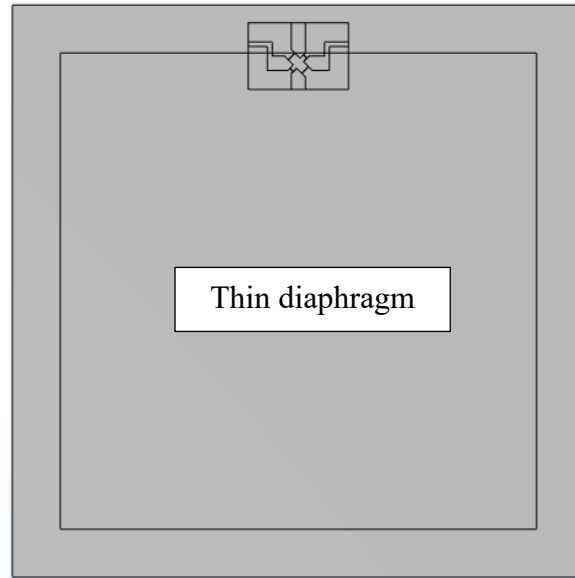


Figure 14: Thin diaphragm with dimensions ( $\mu\text{m}$ )

Table 5: Dimensions of diaphragm

| Parameter           | Value                          |
|---------------------|--------------------------------|
| Diaphragm size      | $1000 \times 1000 \mu\text{m}$ |
| Diaphragm thickness | $5\mu\text{m}$                 |
| Frame thickness     | $20\mu\text{m}$                |
| Cavity depth        | $10\mu\text{m}$                |

### 3.1.2 Piezoresistor Layout and Wheatstone Bridge

Each diaphragm includes a p-type piezoresistor arranged in a Wheatstone bridge on the top silicon surface. The resistors are placed along the longitudinal tensile stress region and two are placed along the transverse compressive stress region. These positions are used to obtain the maximum shear stress.[5]

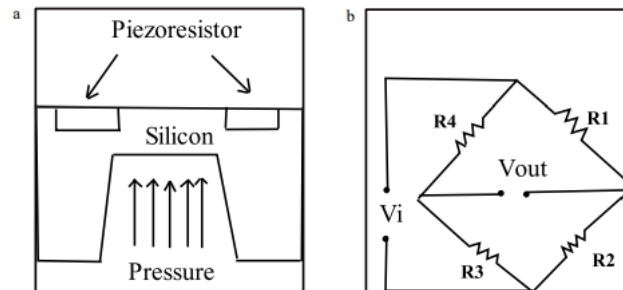


Figure 15: Placement of piezoresistor

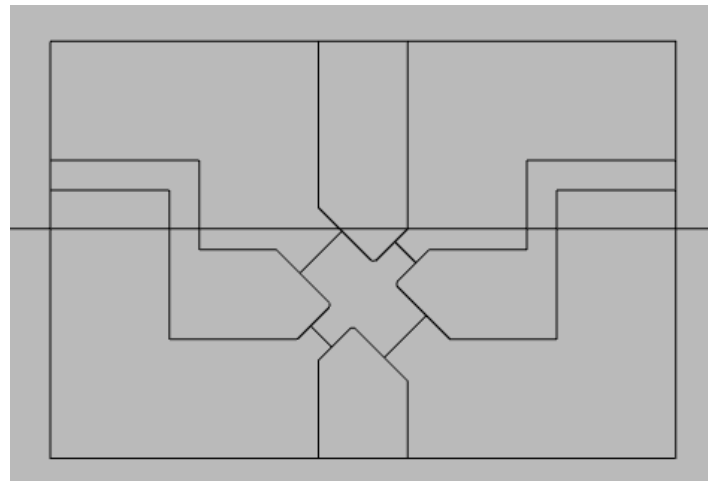


Figure 16: Piezoresistor design

### 3.1.3 Isolation Trench for Independent Movement

Inclusion of an isolation trench surrounding each diaphragm is required because the pressure must be applied for an sensor array therefore for the deflection of the membrane it is required to create a some kind of floating island structure that it can move freely without any mechanical interference. Also, this ensure interference between nearby sensors as well. This shows in the Figure 17.

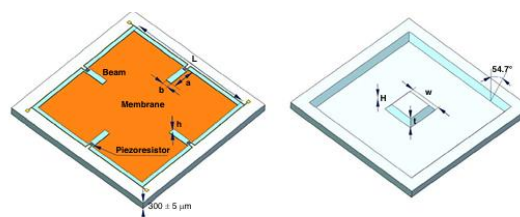


Figure 17: Isolation Trench

### 3.2 Sensor Array Layout for Plantar Pressure Mapping

To capture detailed gait information, a 15-sensor MEMS array should be designed based on plantar pressure distribution maps shown below.

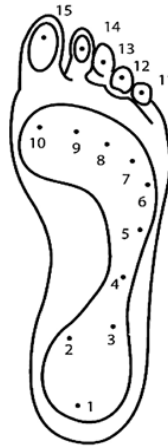


Figure 18: Sensor placing position

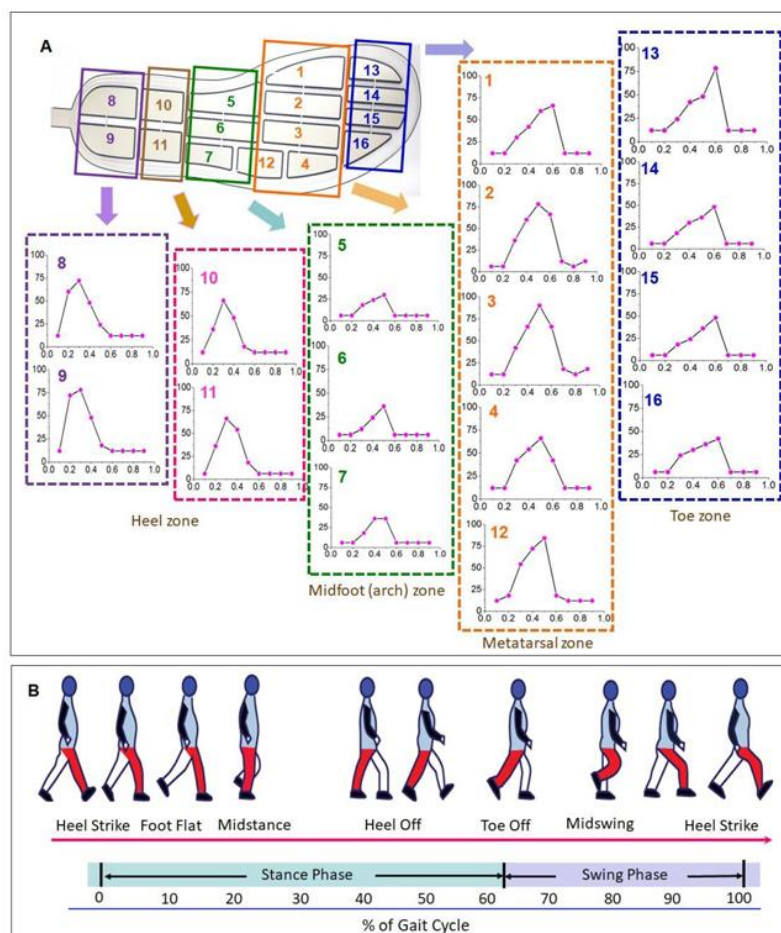


Figure 19: Pressure distribution at the gait cycle[10]

## Sensor Placement Strategy

Sensors are positioned at anatomical high-pressure regions:

- Heel (2 sensors)
- Midfoot (3 sensors)
- Forefoot/metatarsal heads (5 sensors)
- Toes region (5 sensors)

This placement mirrors clinically established foot-loading zones and provides real-time tracking of Center of Pressure (COP), gait phase timing, and posture stability.

The design should be a pressure sensor array. The array must be designed such that all the sensor arrays must have the ability to move their diaphragms when pressure is applied on foot at the gait cycle. Therefore, the diaphragms must separate from an isolation trench, and it must be fixed for the foot sole from a rigid part. Then during the gait cycle pressure is applied in the rigid part to diaphragm to the relevant pressure sensor array and abnormal conditions can be identified.

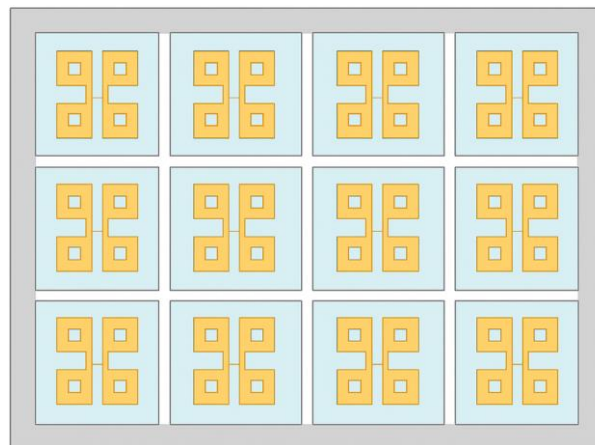


Figure 20: Concept design of the array

Therefore, the final design was done for a single pressure sensor only which is as follows.

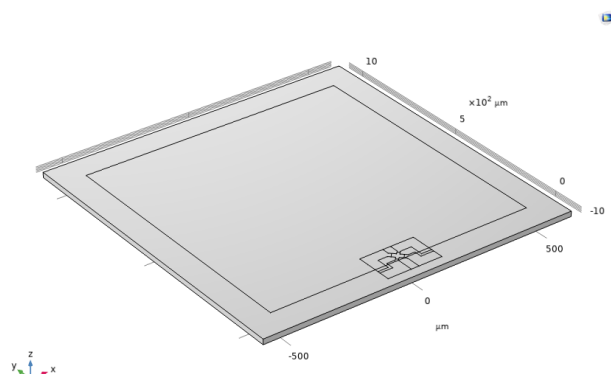


Figure 21: Single sensor design



## 4 Working Principle

The proposed sensor operates based on the piezoresistive effect in a silicon micromachined diaphragm, where externally applied pressure causes diaphragm deformation, generating mechanical stress that modulates the resistance of diffused silicon piezoresistors. These resistors are electrically connected in a Wheatstone bridge to produce a measurable output voltage proportional to the applied pressure.

### 4.1 Pressure Loading and Diaphragm Deflection

When external pressure from the foot is transmitted through the PDMS protective layer, it acts as a uniform pressure load on the MEMS diaphragm.

### 4.2 Stress Distribution and Piezoresistor Placement

As the diaphragm deflects downward, it generates a non-uniform stress distribution,

- Maximum tensile and compressive stresses occur along the diaphragm edges
- The center remains nearly unstressed

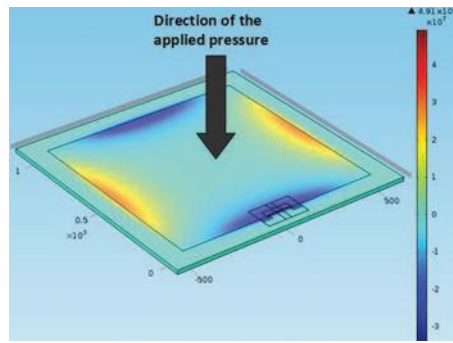


Figure 22: Stress distribution

Therefore, four p-type piezoresistors are placed near the diaphragm edges where stress is highest.

The core electrical quantity in the sensor is the resistivity  $\rho$  of the doped silicon piezoresistors. For a resistor of length  $l$ , cross-sectional area  $A$  and resistivity  $\rho$ , [5]

$$R = \rho \frac{l}{A} [5]$$

If the resistor is strained, its length, cross-section, and resistivity may all change, giving the general relation

$$\frac{\Delta R}{R} = \frac{\Delta l}{l} - \frac{\Delta A}{A} + \frac{\Delta \rho}{\rho} [5]$$

For metals, the geometric terms dominate. In semiconductors like silicon, however, the change in resistivity caused by stress is typically 50–100 times larger than the geometric

contribution.[5] This stress-induced resistivity change is the piezoresistive effect, and we can neglect the geometric terms so that

$$\frac{\Delta R}{R} \approx \frac{\Delta \rho}{\rho} [5]$$

The piezoresistive effect is described by

$$\frac{\Delta \rho}{\rho} = \pi \sigma [5]$$

where  $\sigma$  is the applied stress and  $\pi$  is the piezoresistive coefficient of the material. For a resistor oriented along a particular crystallographic direction, stress can be resolved into longitudinal ( $\sigma_l$ ) and transverse ( $\sigma_t$ ) components, giving

The fractional change in resistance is given by the piezoresistive equation,

$$\frac{\Delta R}{R} = \pi_l \sigma_l + \pi_t \sigma_t [5]$$

where:

- $\pi_l, \pi_t$  = longitudinal & transverse piezoresistive coefficients
- $\sigma_l, \sigma_t$  = stresses along the resistor orientation

This converts mechanical stress into electrical resistance change.

### 4.3 Wheatstone Bridge Conversion

The four piezoresistors form a full Wheatstone bridge, which converts the small resistance variations into a differential voltage,

$$V_{\text{out}} = \left( \frac{\Delta R}{R} \right) V_{\text{exc}}$$

Advantages of the bridge

- Doubles mechanical sensitivity
- Cancels temperature variations
- Produces a strong, linear pressure-to-voltage transfer function

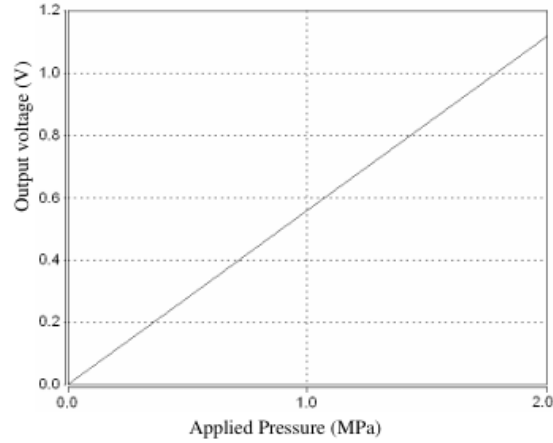


Figure 23: Output voltage vs Pressure

## 5 Material Selection

### 5.1 Silicon as the Structural and Sensing Material

The material that is the backbone to MEMS pressure sensors is silicon, due to its outstanding mechanical and electrical characteristics. It has a large elastic modulus of about 166 GPa, which gives it a great structural rigidity as well as the capability to deform as expected with the loading. It has a clear anisotropic crystal behavior, making the stress and strain very easy to model, which is essential in diaphragm-based sensing. Silicon, especially doped and especially with p-type impurities, has large piezoresistive coefficients hence being very sensitive to even small mechanical forces. Moreover, silicon has an excellent repeatability, low hysteresis and near absence of mechanical creep when subjected to a static and dynamic load. All these features, make single-crystal silicon diaphragms the best choice in plantar pressure sensors where forces may be up to 1 MPa and where stability during the whole lifetime, accuracy, and reliability are the main considerations. The fact that silicon is compatible with CMOS manufacturing and conventional micromachining procedures also makes it more substantial in the design of high-performance MEMS pressure sensors.[2] [5]

### 5.2 n-Type Silicon (Lightly Doped) – Diaphragm Material

| »                                   | Property                               | Variable   | Value         | Unit              | Property group              |
|-------------------------------------|----------------------------------------|------------|---------------|-------------------|-----------------------------|
| <input checked="" type="checkbox"/> | Density                                | rho        | 2330[kg/...   | kg/m <sup>3</sup> | Basic                       |
| <input checked="" type="checkbox"/> | Elasticity matrix, Voigt notation      | {DVo11...  | {166[GPa...   | Pa                | Anisotropic, Voigt notation |
|                                     | Relative permittivity                  | epsilon... | 4.5           | 1                 | Basic                       |
|                                     | Electrical conductivity                | sigma_...  | sigma0(n...   | S/m               | Basic                       |
|                                     | Elasticity matrix                      | {D11,...   | {166[GPa...   | Pa                | Anisotropic                 |
|                                     | Loss factor for elasticity matrix D    | eta_D_i... | 0             | 1                 | Anisotropic                 |
|                                     | Loss factor for elasticity matrix D... | eta_DV...  | 0             | 1                 | Anisotropic, Voigt notation |
|                                     | Piezoresistive coupling matrix, V...   | {Pi11,...  | {-102.2e-...  | Ω·m/Pa            | Piezoresistance form        |
|                                     | Elastoresistive coupling matrix, V...  | {ml11,...  | {-101.4/si... | Ω·m               | Elastoresistance form       |

Figure 24: Material properties for the diaphragm material

Table 6: Material selection for the diaphragm

|                            | Advantages                                                                                                                  | Disadvantages                                                                                                  |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
| Single-crystal Si (n-type) | High Young's modulus, no creep; extremely low hysteresis, well-known anisotropic properties, excellent batch micromachining | requires proper thickness design, rigid needs soft interface above                                             |
| Polysilicon                | CMOS-compatible, can be deposited on many substrates                                                                        | Higher internal stress, more variability, lower mechanical quality than bulk Si                                |
| Glass                      | Good chemical stability, transparent                                                                                        | Much lower piezoresistive effect , bonding to Si adds process complexity                                       |
| Metals                     | High toughness, ductile                                                                                                     | Not compatible with standard MEMS micromachining, residual plastic deformation and creep, poor miniaturization |

Reasons for selection of n-type Silicon for the diaphragm are as follows,

High stiffness and brittle strength allow the membrane to withstand MPa-level plantar pressure.

Anisotropic elasticity matrix accurately models crystal-dependent deflection characteristics.

Low electrical conductivity minimizes parasitic currents, ensuring that only the doped piezo resistors contribute to signal generation.

Stable thermal and mechanical performance, essential for wearable devices subject to temperature variations

### 5.3 p-Type Silicon (Lightly Doped) – Piezoresistor Material

| »                                   | Property                               | Variable   | Value        | Unit                            | Property group              |
|-------------------------------------|----------------------------------------|------------|--------------|---------------------------------|-----------------------------|
| <input checked="" type="checkbox"/> | Relative permittivity                  | epsilon... | 4.5          | 1                               | Basic                       |
| <input checked="" type="checkbox"/> | Electrical conductivity                | sigma_...  | sigma0(n...  | S/m                             | Basic                       |
| <input checked="" type="checkbox"/> | Piezoresistive coupling matrix, V...   | {Pi11,...  | {6.6e-11[... | $\Omega\cdot\text{m}/\text{Pa}$ | Piezoresistance form        |
|                                     | Density                                | rho        | 2330[kg/...  | $\text{kg}/\text{m}^3$          | Basic                       |
|                                     | Elasticity matrix                      | {D11,...   | {166[GPa...  | Pa                              | Anisotropic                 |
|                                     | Loss factor for elasticity matrix D    | eta_D_i... | 0            | 1                               | Anisotropic                 |
|                                     | Elasticity matrix, Voigt notation      | {DVo11...  | {166[GPa...  | Pa                              | Anisotropic, Voigt notation |
|                                     | Loss factor for elasticity matrix D... | eta_DV...  | 0            | 1                               | Anisotropic, Voigt notation |
|                                     | Elastoresistive coupling matrix, V...  | {ml11,...  | {9.6/sigm... | $\Omega\cdot\text{m}$           | Elastoresistance form       |

Table 7: Material selection for the piezoresistor

|                          | Advantages                                                                                                                                 | Disadvantages                                                                                        |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| p-type single-crystal Si | Very high piezoresistive coefficients, 50–100× more sensitive than geometric strain, low hysteresis, can be precisely positioned and doped | Requires doping steps and proper temperature compensation                                            |
| Polysilicon resistors    | Can be deposited after diaphragm formation                                                                                                 | Higher noise, lower gauge factor, significant process variation                                      |
| Metal strain gauges      | Simple deposition and patterning                                                                                                           | Low gauge factor, poor miniaturization for 10–15 $\mu\text{m}$ diaphragms, adhesion and creep issues |
| Polymer FSR inks         | Highly flexible, low cost                                                                                                                  | Strong nonlinearity, drift, and hysteresis, unsuitable for accurate MEMS                             |

Reasons for selection are as follows,

p-Type silicon offers the strongest piezoresistive effect

High sensitivity to mechanical stress makes p-type resistors ideal for micro-pressure sensing.

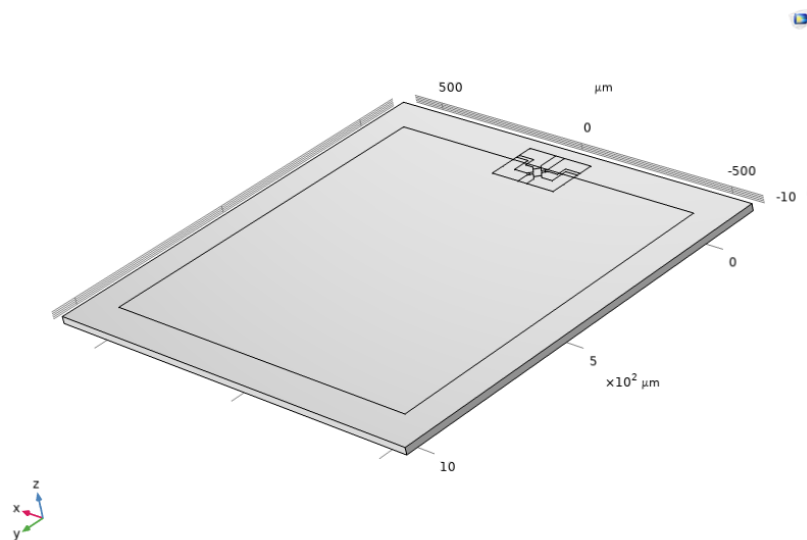
Better thermal stability than polysilicon or metal strain gauges.

Compatibility with diffusion and ion implantation processes, commonly used in MEMS sensor fabrication.

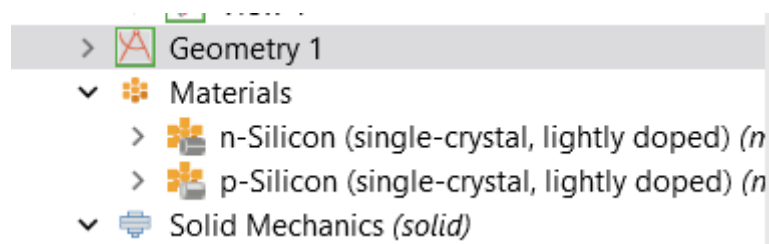
## 6 Simulation and Results

The mechanical deformation, stress distribution, and piezoresistive electrical behavior of the proposed MEMS pressure sensor were studied with the Finite element simulation software, COMSOL Multiphysics. The simulations were intended to confirm the deflection of the diaphragm with the plantar loading, the high-stress areas in which the piezoresistor will be placed optimally, the shear stress fields that will be experienced when using the Wheatstone bridge, the voltage output that would be expected in realistic conditions of the foot-pressure, and the linearity and sensitivity of the design before fabrication. The modelling framework was based on the example of the COMSOL Application Library, Piezoresistive Pressure Sensor, although modified to fit the modified diaphragm geometry, material configuration and the increased pressure range necessary to accurately measure the plantar pressure.

### 6.1 Geometry Setup



### 6.2 Material Assignment

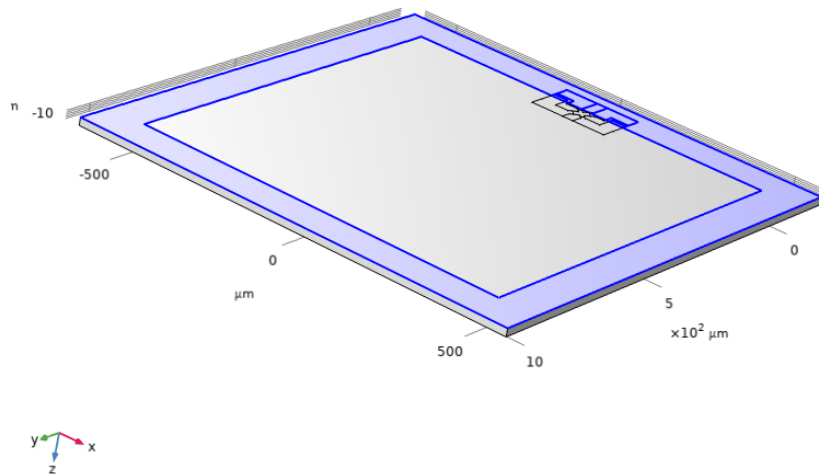


Assign materials for the diaphragm and the piezoresistive sensor.

## 6.3 Physics Setup

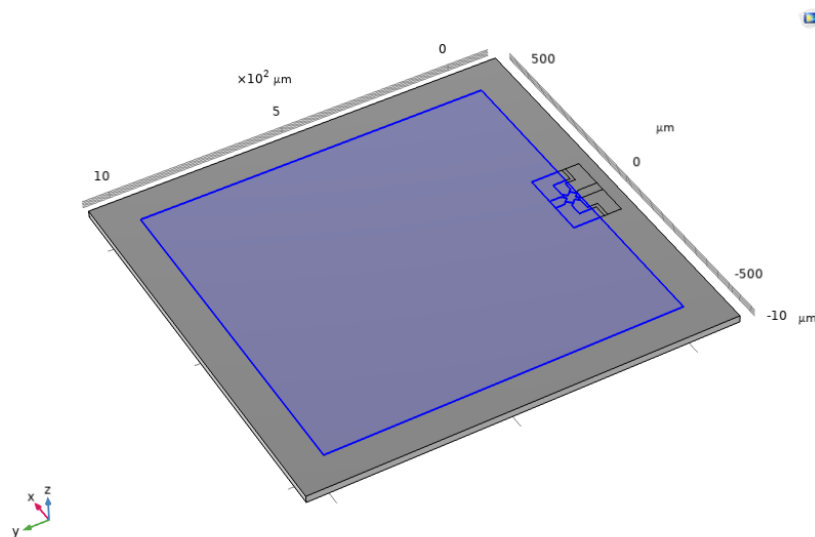
### 6.3.1 Solid mechanics module

All diaphragm edges are fixed to represent the rigid silicon frame.



The applied plantar pressure. State pressure values used is 100 kPa.

The applied pressure is taken from the data values in different gait positions of foot. The simulation will be done for different pressure values to check the sensor design.

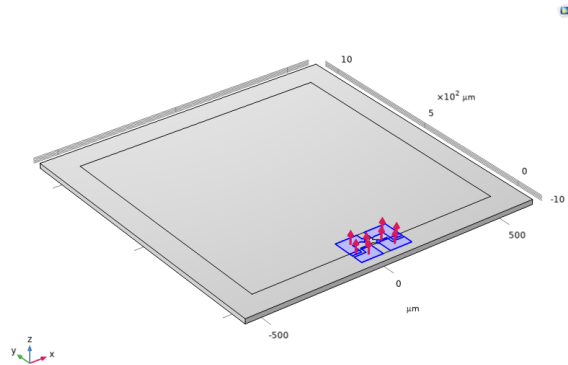


### 6.3.2 Electric currents in Shells

This module handles all electrical behavior of the piezoresistor.

#### Conductive Shell 1

- Represents the thin doped piezoresistor layer.
- Explain that this is a 2D electrical layer on top of a 3D solid.

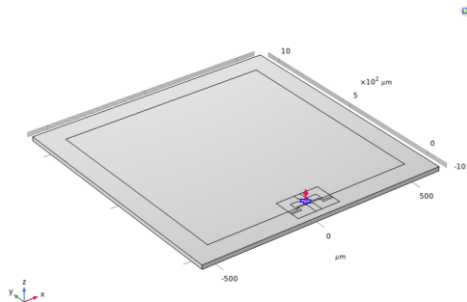


### Electric Insulation 1

- Ensures no unwanted current leakage through surfaces not connected to terminals.

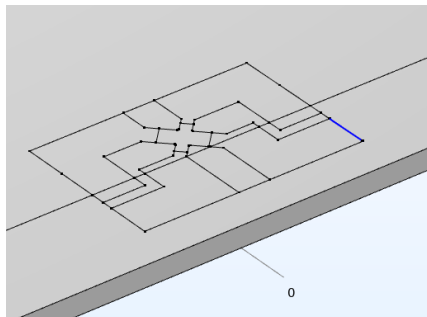
### Piezoresistive Shell 1

- The key multiphysics coupling.
- Converts mechanical stress into electrical resistance change.



### Ground (Terminal 0)

- Defines electrical reference node.



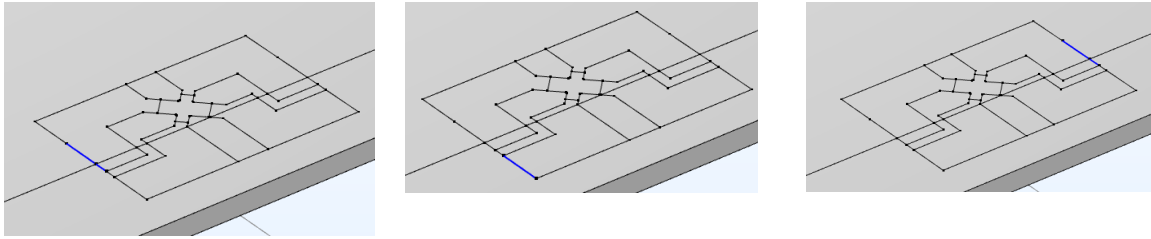
### Terminal 1, Terminal 2, Terminal 3

- Terminal 1 - Voltage input to Wheatstone bridge. At terminal 1 3V is inputted.
- Terminal 2 - Sense terminal

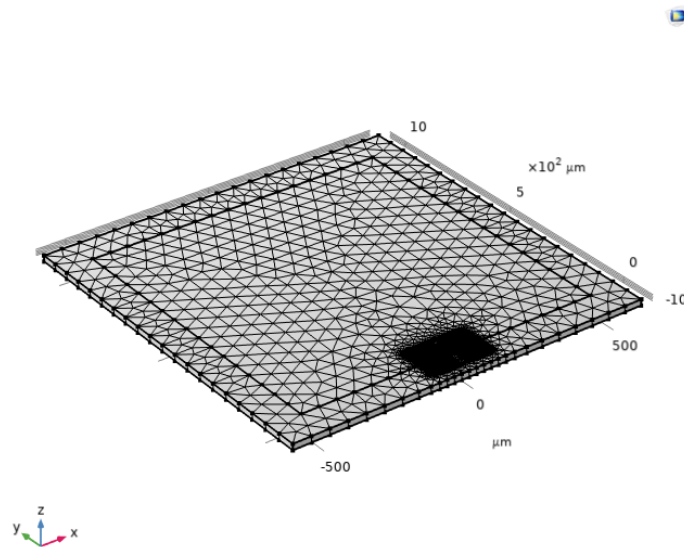


- Terminal 3 - Output terminal

Describe how each terminal corresponds to the Wheatstone bridge nodes.



## 6.4 Meshing



A refined mesh was applied to accurately capture the diaphragm deformation and the stress distribution around the piezoresistors. The silicon diaphragm was meshed with a fine tetrahedral grid, while the piezoresistive shell regions received extra refinement due to their small size and steep electrical and mechanical gradients. This ensured stable convergence of the multiphysics model and reliable results for both displacement and voltage output. refined mesh was applied to accurately capture the diaphragm deformation and the stress distribution around the piezoresistors.

### Complete mesh

Mesh vertices: 7544

Element type: All elements ▼

Prisms: 7455

Triangles: 14910

Quads: 782

Edge elements: 1626

Vertex elements: 124

#### — Domain element statistics —

Number of elements: 7455

Minimum element quality: 0.3733

Average element quality: 0.8077

Element volume ratio: 3.774E-6

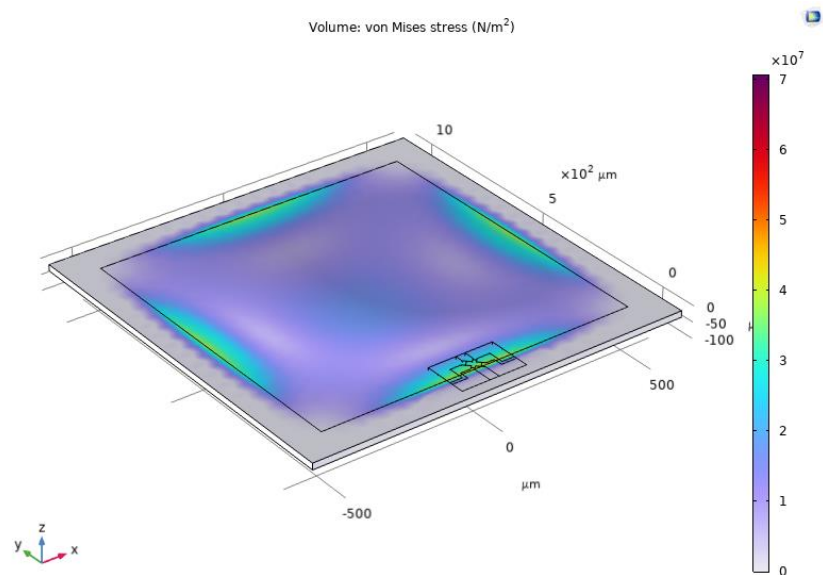
Mesh volume: 2.88E7  $\mu\text{m}^3$

### Element Quality Histogram

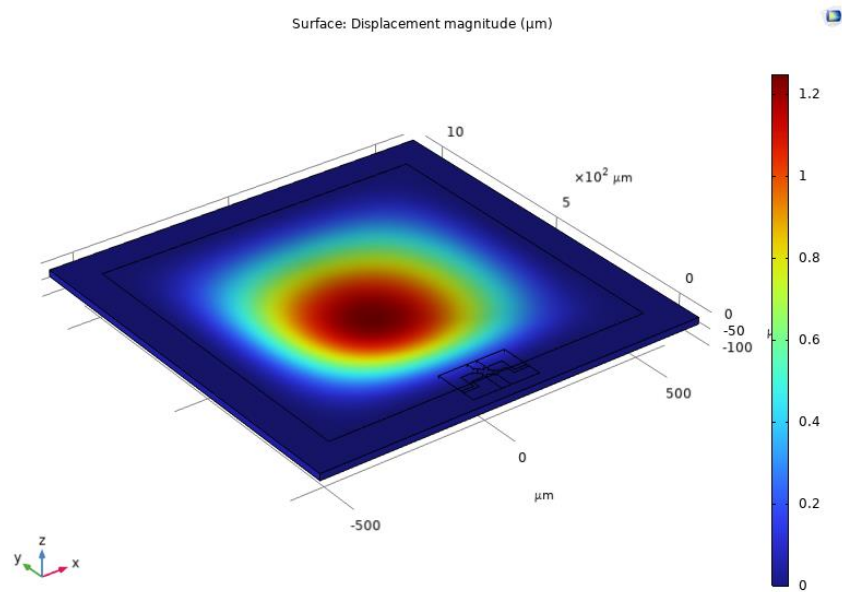


## 6.5 Results

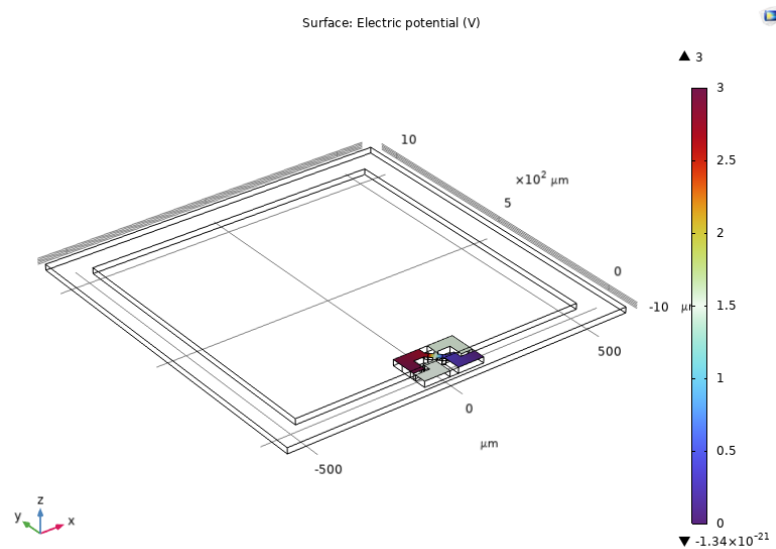
### 6.5.1 Stress Distribution at 100kPa



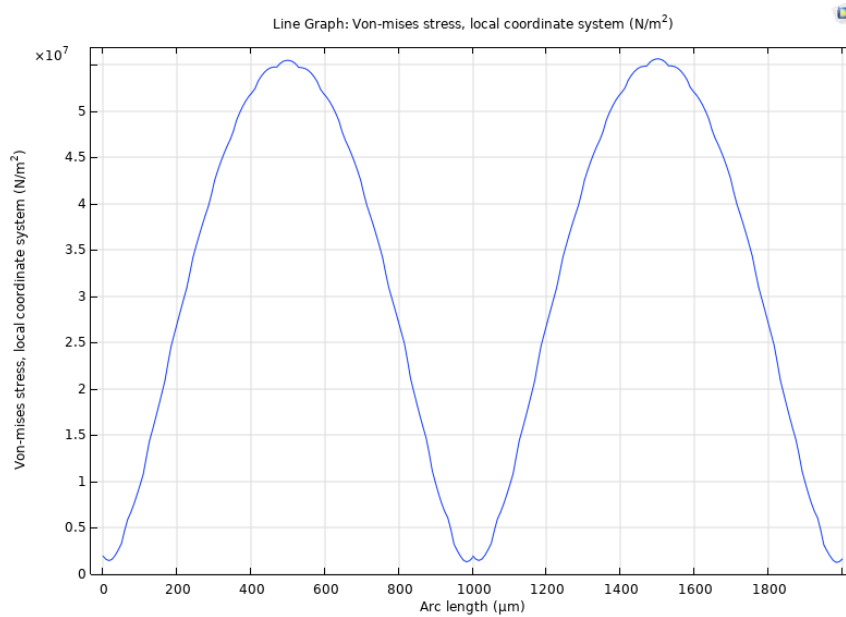
### 6.5.2 Displacement Field



### 6.5.3 Electric Potential



#### 6.5.4 Stress distribution along the diaphragm edge



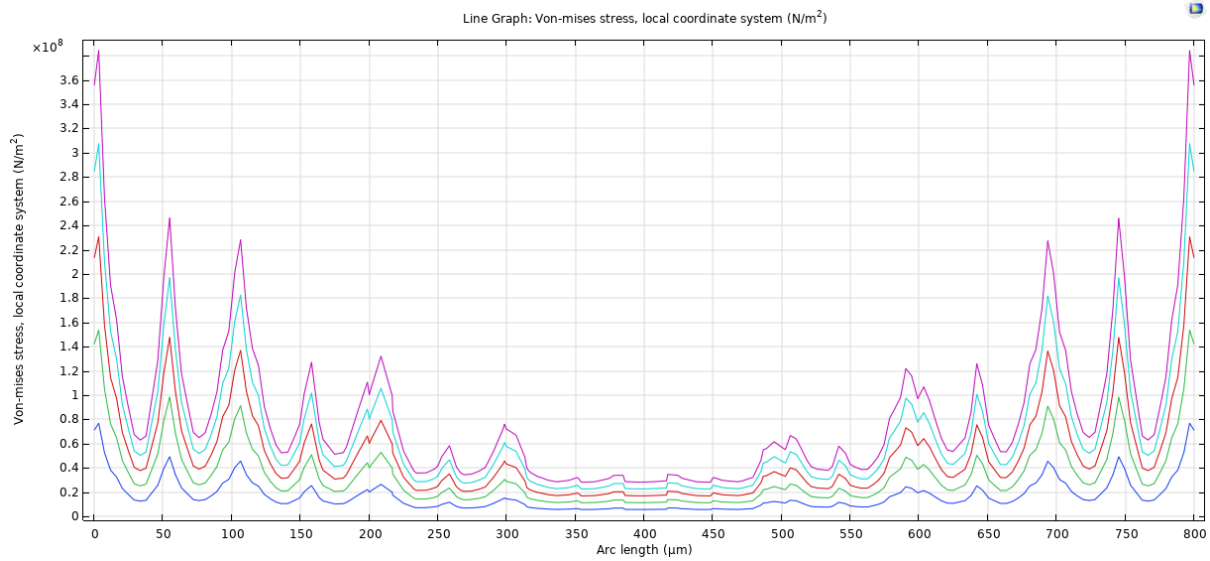
This shows how the mechanical stress varies along the arc length of the selected diaphragm edge when a pressure load is applied.

### 6.5.5 Output Voltage

|                       |                    |  |
|-----------------------|--------------------|--|
| Terminal current (mA) | Device Output (mV) |  |
| 5.9213                | -51.402            |  |

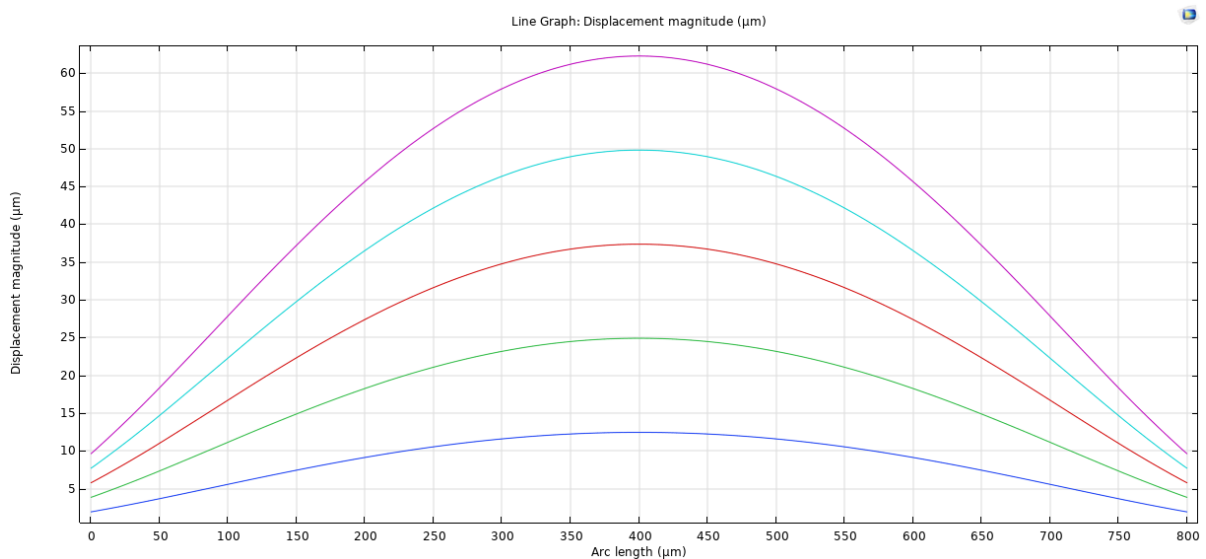
Output voltage when 100kPa voltage is applied.

### 6.5.5 Stress analysis at different plantar pressure values



The von Mises stress plot illustrates the stress distribution along the diaphragm boundary for increasing pressure loads. Stress is highest near the diaphragm edges and corners consistent with analytical plate-bending theory and increases proportionally with applied load. Peak stresses remain below the fracture threshold of single-crystal silicon (7 G Pa), demonstrating safe operating conditions even at the upper design limit of 2 MPa. These stress peaks are also located exactly where the piezo resistors are positioned, validating that the sensor layout maximizes strain sensitivity and electrical response.

### 6.5.6 Displacement at different pressure values



The displacement curves show how the silicon diaphragm deflects along a radial line as plantar pressure increases from 0.5 MPa to 2 MPa. As expected for a square diaphragm, the displacement profile takes a smooth parabolic shape, reaching its maximum at the center. Higher applied pressure produces proportionally larger deflection, with the peak displacement increasing from 12  $\mu\text{m}$  (0.5 MPa) to over 60  $\mu\text{m}$  (2 MPa). This confirms that the diaphragm operates within a predictable, quasi-linear deformation regime suitable for piezoresistive sensing. Importantly, the maximum deflection values remain well below the typical cavity depth, ensuring that no bottoming-out or nonlinear contact occurs during gait-level loading.

## 6.6 Modal Analysis

### Eigenfrequency (Hz)

2.6747E5

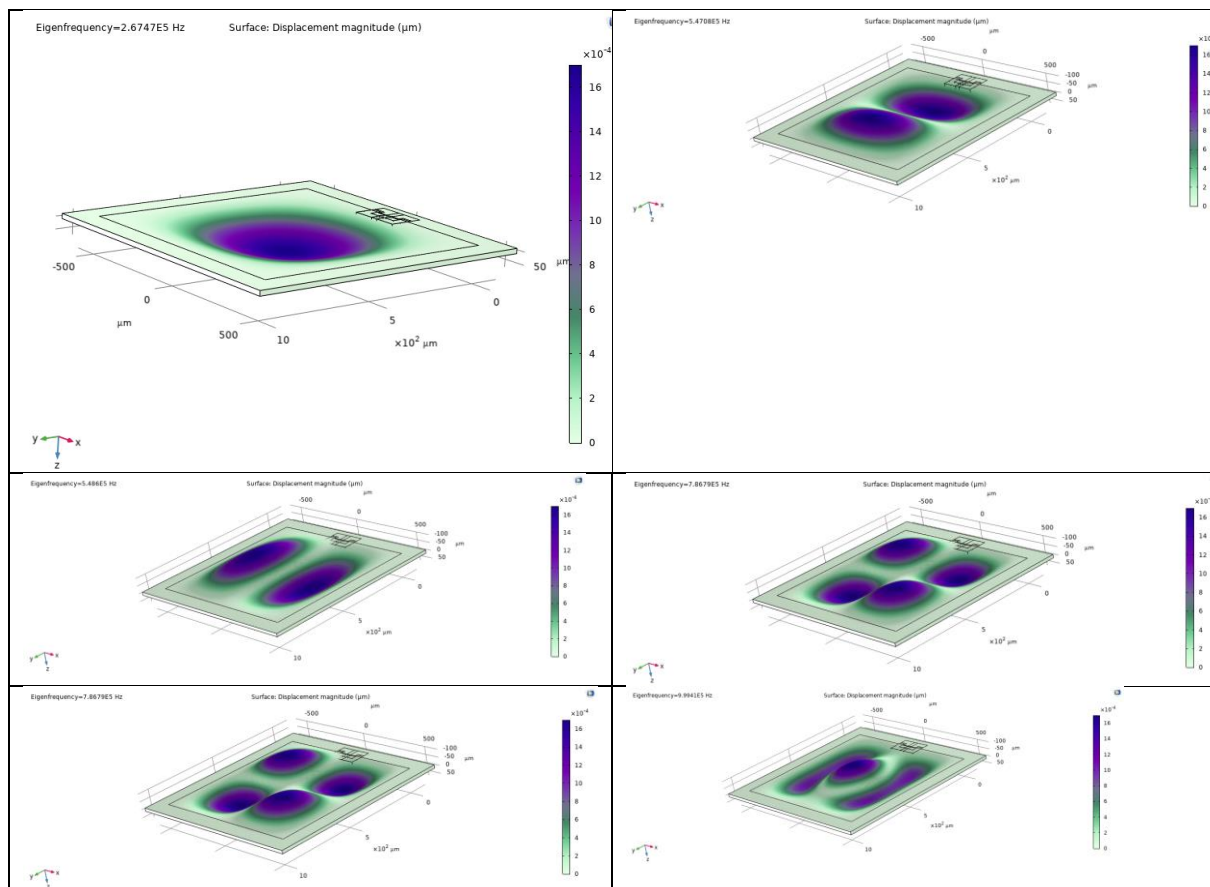
5.4708E5

5.4860E5

7.8679E5

9.9297E5

9.9941E5



The modal analysis of the diaphragm reveals its natural frequencies under no applied pressure. The first five eigenfrequencies are shown, ranging from approximately 547 kHz to 9.99 MHz, with the highest displacement occurring at the diaphragm center. These frequencies represent the diaphragm's vibrational modes, which do not interfere with the operational pressure range of 0–2 MPa. The diaphragm's first eigenfrequency (~547 kHz) is well above the operational frequencies, indicating that mechanical vibrations during gait loading will not affect the sensor's accuracy.

## **7 Fabrication Process**

### **Starting wafer**

Begin with a 20  $\mu\text{m}$  thick, <100> p-type silicon wafer which is double-side polished. This will be thinned locally to obtain the diaphragm while leaving a thick frame for mechanical support.

### **Initial thermal oxidation**

Grow a  $\text{SiO}_2$  layer about 2  $\mu\text{m}$  on both sides of the wafer using thermal oxidation. This oxide is used as an etch mask for subsequent KOH/TMAH silicon etching and as an electrical isolation layer.

### **Front-side photolithography**

Coat the front side with the photoresist and protect the back side with wax then use a front-side mask to pattern the shallow cavity area where the diaphragm thickness will be reduced. Then expose and develop the photoresist to open windows in the resist above the diaphragm region.

### **Front-side oxide etching**

Use buffered HF or another oxide etchant to etch the exposed  $\text{SiO}_2$ , opening the diaphragm pattern down to silicon. Then, strip the photoresist and clean the wafer.

### **Front-side KOH etch and re-oxidation (diaphragm pre-thinning)**

Perform a shallow KOH etch through the oxide opening to create a recess of about 5  $\mu\text{m}$  depth which defines the final diaphragm thickness. After etching, remove the wax, clean the wafer again, and grow a new thermal oxide on the front side to re-passivate the surface.

### **Back-side photolithography (diaphragm window)**

Spin-coat the photoresist on the back side of the wafer. Use a backside alignment mask to pattern the large diaphragm window on the back oxide, aligned to the shallow cavity formed on the front. Then, expose and develop to open the photoresist over the diaphragm region.

### **Back-side oxide etching**

Etch the exposed backside  $\text{SiO}_2$  to open the **silicon window** that will be used for bulk etching. Strip the photoresist and clean the wafer. At this point, Si is exposed on the backside only in the diaphragm region.

### **Bulk silicon etching to form membrane**

Perform deep anisotropic etching from the backside using KOH and/or TMAH. Etching stops near the previously defined 5  $\mu\text{m}$  recess, leaving a thin silicon diaphragm under the front-side cavity. This defines the mechanical membrane that will deflect under plantar pressure.

### **Front-side photolithography for piezoresistor regions**

On the diaphragm side, deposit or maintain a thin oxide and coat photoresist again. Use a mask to pattern the piezoresistor shapes in the high-stress regions around the diaphragm edges. Etch the oxide in these windows to expose silicon where the diffused resistors will be formed.

### **Dopant diffusion to form piezoresistors**

Perform phosphorus diffusion at about 900 °C in the opened regions to create n-type piezoresistors inside the p-type diaphragm. The diffused regions form the four active resistors of the Wheatstone bridge and are aligned to the simulated high-stress locations. Strip any remaining photoresist and clean the wafer.

### **Metallization and patterning**

Deposit a thin layer of aluminum (Al) over the front side using thermal evaporation or sputtering. Apply photoresist and use a metal mask to pattern,

- Contact pads on top of each piezoresistor
- Interconnect lines forming Wheatstone bridge.

Etch away unwanted Al, leaving the patterned metal network. Anneal the wafer to form low resistance ohmic contacts between Al and the diffused regions.

### **Final device layout and array formation**

After metallization, the front side now shows the full bridge layout with resistors  $R_1$ – $R_4$  and bonding pads.

The wafer is diced into individual sensor chips. Multiple chips can then be arranged into a 15-node plantar pressure array on a flexible PCB for integration into the insole.

Subsequent packaging protects the diaphragm while allowing pressure transmission from the PDMS or insole layer to the MEMS membrane.



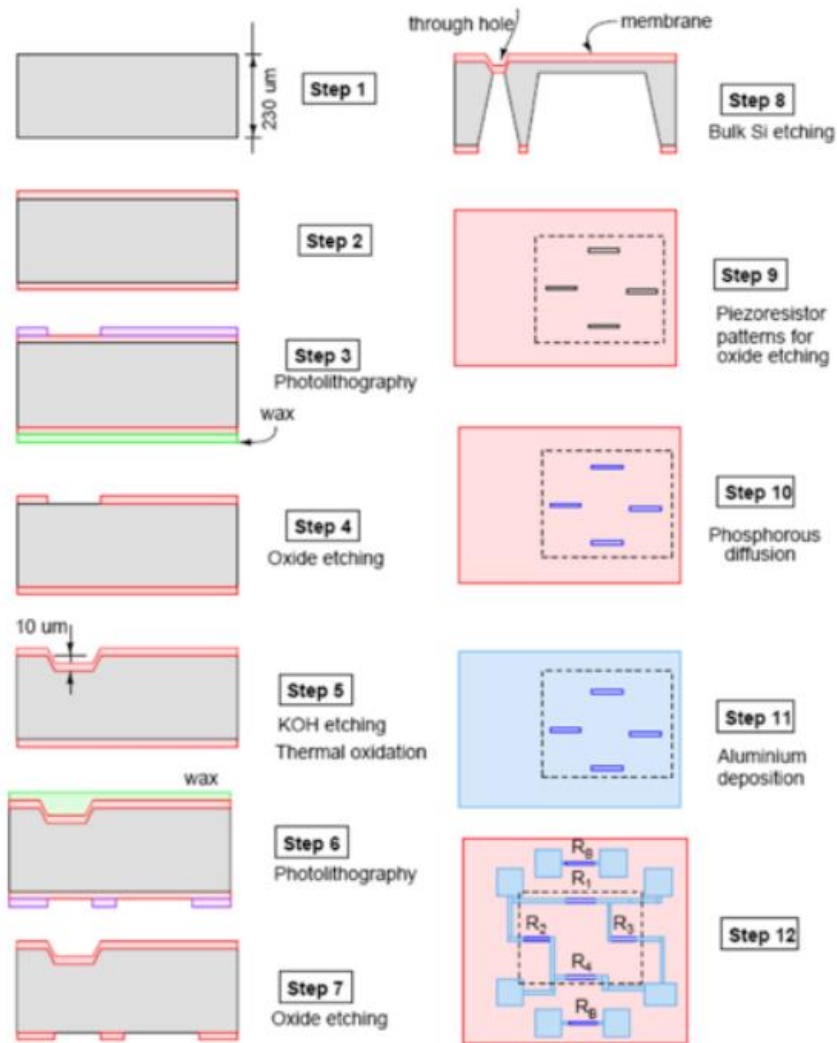


Figure 25: Fabrication Steps[11]

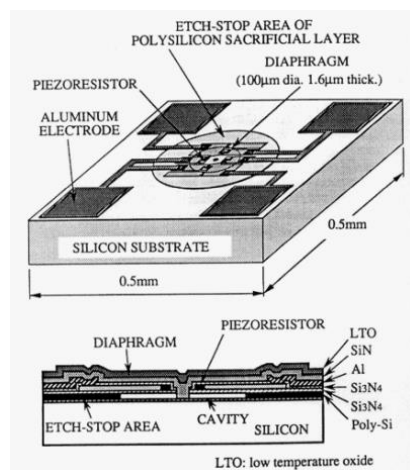


Figure 26: Schematic and cross-sectional view of micromachined piezoresistive pressure sensor[11]

## 8 Conclusion

The project has been able to design and simulate a MEMS piezoresistive pressure sensor array that can be used to monitor posture and gait in real-time during wear by shoes. The sensor proposed makes use of a silicon diaphragm and piezoresistors of the p-type and is strategically positioned at high-stress areas to gain maximal sensitivity. The simulation outcomes, carried out with the help of the COMSOL Multiphysics, confirmed the mechanical diaphragm deflection to the realistic plantar pressures (0 -2 MPa). The distribution of stress was also in agreement with the theory whereby the extreme stress concentrations were observed at the edges of the diaphragm, which validated the best location of the piezoresistive elements.

The parametric sweep experiments revealed that there was a linear dependence between applied pressure and sensor output and thus the sensor is suitable in dynamic gait analysis. The von Mises stress analysis was used to demonstrate that the sensor design is mechanically safe, and the maximum stress is significantly lower than the fracture strength of silicon. Also the deflection analysis indicated that the diaphragm deflects in the linear range which also confirms the design of the diaphragm as accurate and reliable pressure sensing.

The design also benefits certain limitations common to the existing plantar pressure measurement systems including hysteresis, nonlinearity, and drift due to the high-quality silicon material with low creep and high repeatability. Moreover, piezoresistive MEMS technology is sensitive and can be integrated thus making it appropriate in detecting small changes in plantar pressure which is of great relevance in early detection of foot abnormalities such as diabetic foot ulcers.

Finally, the MEMS piezoresistive pressure sensor array developed within the framework of the current project has a strong potential in being used as a wearable sensor, as it represents a powerful, sensitive, and affordable system to measure gait and posture in a healthcare and non-clinical environment. The future research is directed at streamlining the fabrication process, dealing with the long-term drift, and strengthening the integration of the sensor into commercial footwear in order to deploy it into the real world.

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| Sensor Type         | Sensitivity | Static Pressure Capability | Power Requirement | Complexity | Suitability for Plantar Pressure Measurement |
|---------------------|-------------|----------------------------|-------------------|------------|----------------------------------------------|
| Capacitive MEMS     | High        | Medium                     | Low               | High       | Moderate                                     |
| Piezoresistive MEMS | Very High   | High                       | Moderate          | Low        | Excellent                                    |
| Piezoelectric MEMS  | High        | Low                        | None              | Moderate   | Poor                                         |
| Optical MEMS        | Very High   | Medium                     | High              | Very High  | Poor                                         |